

Nur Mohammad Ali

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❖ Research Interest

My research focuses on AI-driven decision support for industrial systems to improve U.S. supply chain resilience, advanced manufacturing productivity, and energy-efficient operations. My work combines machine learning and large language models (LLMs) to develop deployable models and user-centered workflows that convert operational and sensor data into actionable decisions.

Programming & Tools: Python (Pandas, NumPy, Matplotlib, Scikit-learn), SQL (MySQL), Tableau, SAP ERP, Power BI.

❖ Education

- **Doctor of Engineering, Industrial Engineering**, Lamar University 2025 - Present
Research Focus: Large Language Models (LLM), Deep Learning,
Current project: Leveraging deep learning and large language models for advance vehicle fault diagnosis
- **MS in Industrial Engineering**, Lamar University, USA 2023 – 2024
Coursework: Additive manufacturing, Machine learning, data analysis

❖ Research work

- **Nur Mohammad Ali**, Md Kamrul Hasan Tuhin, Rezwanul Ashraf Ruddro, Md Emon Ahmed, MD Shafiqul Alam, Nowrin Sharmin, and Jayanta Bhusan Deb, "*A Fuzzy Inference System for Predicting Air Traffic Demand based on Socioeconomic Drivers*," Saudi Journal of Engineering and Technology, vol. 9, issue 8, pp. 377-388, 2024
- Jayanta Bhusan Deb, Shilpa Chowdhury, and **Nur Mohammad Ali**, "*An investigation of the ensemble machine learning techniques for predicting mechanical properties of printed parts in additive manufacturing*," Decision Analytics Journal, vol. 12, 100492, 2024
- Fakir Sheik Zihad, Mrinmoy Sen, Fardin Sabahat Khan, Md Siam Taluckder, Fozlur Rayhan, Sohel Rana, and **Nur Mohammad Ali**, "*Enhancement Study on Lumen Depreciation & Thermal Management System in Commercial LEDs Bulb from a Product Lifecycle Perspective*," in Proceedings of the 5th African International Conference on Industrial Engineering and Operations Management, 2024

Submitted and under review

- U.S. Path to Industry 4.0: Reassessing Supply Chain Digitalization and AI for Industrial Sustainability; Journal: ELSEVIER

❖ **Professional Experience**

Assistant Supply Chain Manager, Hyundai Bangladesh Feb. 2022 – Jan 2023

- Engineered supply chain workflows to integrate 15 new vendors, utilizing process mapping to reduce material lead times.
- Conducted statistical analysis on inventory data using Python and SQL to identify variance root causes and reduce stock discrepancies
- Optimized SKD (Semi-Knocked Down) planning parameters within a \$55K monthly budget, achieving a 15% cost reduction through demand smoothing

Senior Supply Chain Executive, Toyota Bangladesh June 2018 – Jan 2022

- Designed and implemented a Just-In-Time (JIT) inventory model, increasing parts availability by 20% while minimizing carrying costs
- Developed deterministic demand forecasting models for 120+ SKUs using historical data analysis, resulting in improved forecast accuracy.
Led the alignment of sales and logistics data to streamline operations for a 20-person cross-functional team

Procurement Officer, Walton Group Feb 2016 – May 2018

- Analyzed international logistics data to optimize 3PL performance for spare parts shipment
- Managed procurement logistics for a \$52,000 USD budget, ensuring critical path alignment for new product launches

❖ **Extra-curricular activities**

- *Beaumont School Visit:* In 2025, I visited a local school in Beaumont and spoke with students about international experiences and educational goals, encouraging them to explore education with confidence and kindness.
- *Digital Empowerment Initiative:* In 2024, trained 32+ senior citizens in Houston Chinese community center on essential technologies, including AI tools like ChatGPT, digital safety for identifying fake content, and navigation with Google Maps.
- *Volunteering at the Houston Food Bank:* In **2023**, volunteered at the Houston Food Bank by organizing, packing, and distributing food for families in need, contributing through teamwork and community service